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# Lifted Cavity Networks: In-Context Sensor Fusion with Missing Sensors and Shared Noise

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## Abstract

1 Low-cost sensor deployments must estimate a target from whichever sensors work,  
2 after a short calibration, while temperature, humidity and drift perturb several sen-  
3 sors at once. We cast this as in-context probabilistic prediction that must operate  
4 under different missingness patterns and numbers of sensors. For linear-Gaussian  
5 sensors with shared noise, we show that mask-independent scalar Gaussian ex-  
6 perts cannot in general reproduce every pattern’s posterior. We propose *lifted*  
7 *cavity networks*: each observed sensor contributes a rank-one Gaussian site over  
8 the target and a low-dimensional shared nuisance, exact for factor-structured noise  
9 under every pattern once the nuisance dimension reaches the noise rank; necessity  
10 holds for static linear-Gaussian sites under a generic-rank condition. The sites  
11 start from a factor analysis of the incomplete calibration rows and are refined by  
12 a learned, cavity-conditioned re-linearization. On synthetic tasks with exact oracle  
13 and sign-conditioned HMC references, an 8k-parameter lifted cavity network  
14 closes 58% of the gap between the closed form and the HMC reference with two  
15 sensors missing, a pattern unseen in training, against 34% for a 25-times-larger  
16 TabPFN-v2-style transformer trained on the same tasks, and it transfers from five  
17 sensors to sixteen. After fine-tuning on real sensor networks it improves on the  
18 closed forms, but the transformer is better still. As the transformer’s output layer,  
19 lifted sites beat the plain transformer on fresh synthetic tasks (by 0.035 nats with  
20 five sensors, 0.150 with sixteen) and recover most of the real-data gap on a new  
21 target, although non-inferiority to the fine-tuned transformer is not established.  
22 All three evaluation protocols were committed before their test data existed, and  
23 every endpoint is reported.

## 1 Introduction

25 Low-cost sensors now carry much of the world’s environmental, industrial and clinical measurement.  
26 A typical deployment co-locates a few cheap sensors with a reference instrument for a short calibra-  
27 tion period, and from then on predicts the reference quantity from the sensors alone [39]. Three  
28 features make this hard. First, every deployment is different: gains, cross-sensitivities and noise de-  
29 pend on the unit, its placement and its age, and the calibration record is short. Second, sensor noise  
30 is *shared*: temperature, humidity, local emissions and drift move several sensors at once, so sensors  
31 are not conditionally independent given the target. Third, sensors fail, are serviced or are added, so  
32 at prediction time an arbitrary subset is available, often one never seen together during calibration.

33 We study this as *in-context* probabilistic prediction. A network receives a deployment’s labeled cali-  
34 bration rows (the support), with whatever entries are missing, and a query made of an arbitrary subset  
35 of sensor readings, and it returns a predictive distribution for the target. Trained once across many  
36 deployments, such a network adapts to a new one from its calibration rows alone, as conditional  
37 neural processes and prior-fitted networks do for regression and tabular prediction [8, 26, 15].

The problem imposes two structural requirements. The predictor must be *mask-consistent*: defined and accurate for every subset of observed sensors, including subsets absent from training. It must also be *width-agnostic*: usable for any number of sensors. A product of per-sensor experts over the target meets both by construction, because a missing sensor simply contributes no factor [13, 36]. Learned message passing in the style of expectation propagation (EP) refines such experts using context from the other sensors [24, 12].

#### Contributions.

1. **When products of experts fail (§3–4).** Under shared noise, no product of mask-independent experts over the target alone is Bayes-consistent across missingness patterns; the optimal weight of a sensor can even change sign with the other sensors present. Under the generic-rank assumptions of Proposition 3, exact static linear-Gaussian sites require at least  $K$  nuisance dimensions for rank- $K$  shared noise. These results do not exclude mask- or cavity-conditioned scalar sites.
2. **Lifted cavity networks (§4).** Each observed sensor contributes a rank-one Gaussian site over the target and a low-dimensional shared nuisance, and missing sensors are omitted exactly. The sites start from a factor analysis of the incomplete calibration rows and are refined by a small network that re-linearizes each sensor’s response under its cavity.
3. **A pre-registered evaluation (§6)** against exact posteriors, a Hamiltonian Monte Carlo reference conditioned on support-inferred response signs, tuned closed forms and a TabPFN-v2-style transformer trained on the same task distribution, on synthetic panels and two real sensor networks. Three protocols were committed before their test data existed, and every endpoint is reported.

**What we find.** Structure buys sample efficiency and extrapolation, but on real data it costs flexibility.

- On synthetic tasks, the 8k-parameter lifted cavity network beats a 203k-parameter TabPFN-v2-style transformer trained on the same tasks for every number of missing sensors, closing 58% of the HMC reference gap against 34%; its margin grows on 16 sensors after training on five. It loses its edge under strong response nonlinearity and on linear sensors (§6.3).
- On two real sensor networks, the fine-tuned lifted network improves on the closed forms, but a fine-tuned transformer is better, and one pre-registered endpoint fails (§6.5).
- As the transformer’s output layer (§6.6), lifted sites beat the plain transformer on fresh synthetic panels and the lifted network on a new real target, recovering most of the real-data gap (75% for the pre-registered seed, 62% averaged over two); non-inferiority to the fine-tuned transformer is not established.

## 2 Problem setup

A task (deployment)  $\tau \sim p(\tau)$  defines a joint distribution  $p_\tau(x, y)$  over  $P$  sensor readings  $x \in \mathbb{R}^P$  and a scalar target  $y$ . The support  $\mathcal{D} = \{(x_i, m_i, y_i)\}_{i=1}^n$  holds  $n$  labeled rows whose entry  $x_{ij}$  is observed where  $m_{ij} = 1$ . A query is a reading  $x_S$  of an observed subset  $S \subseteq \{1, \dots, P\}$ . The target of prediction is the posterior predictive

$$p(y \mid x_S, \mathcal{D}) = \int p_\tau(y \mid x_S) p(\tau \mid \mathcal{D}, x_S) d\tau, \quad (1)$$

and methods are scored by the negative log-likelihood (NLL) of the held-out target. Two privileged references bound what is possible on synthetic tasks: the *oracle*  $p_\tau(y \mid x_S)$ , which knows the task, and the *Bayes-optimal predictive* (1), which knows only the task prior. Our numerical HMC reference approximates the latter after conditioning on response signs inferred from the support; it is not a certified Bayes-optimal ceiling. Reported gap-closure percentages use the NLL difference between the closed-form FA-Gaussian of §4 and this sign-conditioned HMC reference (the *reference gap*).

An in-context method learns a map  $(\mathcal{D}, x_S) \mapsto q(y)$  by minimizing NLL over tasks drawn from a training distribution [8, 26, 9]. Throughout, models are trained only on queries with *at most one* missing sensor, and evaluated on patterns with up to three missing. Two- and three-sensor deletions are therefore never seen in training.

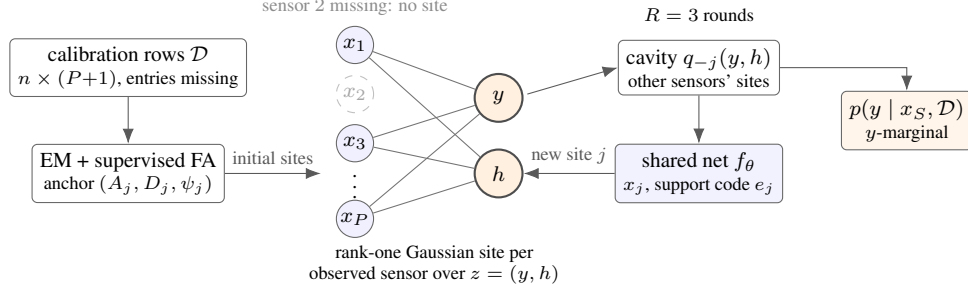


Figure 1: Lifted cavity network. A supervised factor analysis estimated from the incomplete calibration rows gives every sensor a rank-one Gaussian site over the target  $y$  and a shared nuisance  $h$ . Missing sensors contribute no site, so any subset and any number of sensors are handled by omission. For  $R$  rounds, a shared network replaces each observed sensor’s site using its cavity (the posterior from the other sensors) and an encoding of that sensor’s calibration scatter. The prediction is the  $y$ -marginal of the product.

### 3 Scalar experts cannot fuse sensors with shared noise

Consider  $x_j = a_j y + d_j u + \sigma_j \epsilon_j$  with mutually independent  $y, u, \epsilon_j \sim \mathcal{N}(0, 1)$ ,  $\sigma_j > 0$  and  $a_j \neq 0$ . Here  $u$  is a nuisance shared by all sensors. A product-of-experts (PoE) predictor gives each sensor a factor  $t_j(y; x_j)$  and predicts  $q(y | x_S) \propto p(y) \prod_{j \in S} t_j(y; x_j)$ . Omitting missing sensors is what makes the construction mask-consistent.

**Proposition 1** (Scalar experts). *Let each  $t_j$  be Gaussian in  $y$  with positive precision, depend on  $x_j$ , and not depend on  $S$ . Suppose  $q$  equals the Bayes posterior for every reading in each singleton query  $S = \{j\}$ . For a pair  $\{j, k\}$ ,  $q$  equals the Bayes posterior for every reading if and only if  $d_j d_k = 0$ . Moreover,  $q$ ’s pair posterior mean gives  $x_j$  a coefficient with the sign of  $a_j$ , whereas the Bayes coefficient can have the opposite sign.*

The proof (Appendix A) shows that matching single-sensor queries forces  $t_j(y; x_j) \propto p(x_j | y)$ , so  $q$  is the naive-Bayes posterior. For example, take  $a = (1, 1)$ ,  $d = (1, 0.5)$  and  $\sigma = (0.05, 0.05)$ . The Bayes coefficient of  $x_1$  is  $+0.50$  when  $x_1$  is alone and  $-0.96$  in the pair: the second sensor lets the predictor estimate  $u$  and subtract it from the first.

EP [24] refines each factor using its *cavity*, the approximate posterior given the other sensors, and learned EP amortizes the refinement [12, 6, 16]. With a Gaussian over  $y$  alone, however, the cavity summarizes the other sensors by a scalar mean and variance, discarding what they reveal about  $u$ . Proposition 1 does not exclude every scalar-cavity network, but the failure is visible in practice (§6.4).

### 4 Lifted cavity networks

**Lifted sites.** Let  $z = (y, h) \in \mathbb{R}^{1+K}$ , with independent  $h \sim \mathcal{N}(0, I)$  and  $y \sim \mathcal{N}(\mu_y, v_y)$ ,  $v_y > 0$ . Write the prior in natural parameters  $(\Lambda_0, \eta_0)$ . Each observed sensor contributes a rank-one Gaussian site, which corresponds to a linear-Gaussian measurement  $\tilde{x}_j = w_j^\top z + e_j$  with  $e_j \sim \mathcal{N}(0, \psi_j)$ :

$$\Lambda_j = \frac{w_j w_j^\top}{\psi_j}, \quad \eta_j = \frac{w_j o_j}{\psi_j}. \quad (2)$$

The prediction is the  $y$ -marginal of

$$q(z | x_S) = \mathcal{N}(z; \Lambda^{-1} \eta, \Lambda^{-1}), \quad \Lambda = \Lambda_0 + \sum_{j \in S} \Lambda_j, \quad \eta = \eta_0 + \sum_{j \in S} \eta_j. \quad (3)$$

**Proposition 2** (Exactness). *Suppose  $x_j = A_j y + D_j^\top h + c_j + e_j$ , with  $\psi_j > 0$ , mutually independent  $e_j \sim \mathcal{N}(0, \psi_j)$  independent of  $z$ , and the prior above. Set  $w_j = (A_j, D_j)$  and  $o_j = x_j - c_j$ . Then (3) equals  $p(z | x_S)$  for every fixed  $S \subseteq \{1, \dots, P\}$ . Its  $y$ -marginal is Bayes predictive for noninformative observation masks; informative masks require a selection model.*

115 Shared noise of rank  $K$  is absorbed by  $K$  nuisance dimensions, so the obstruction of Proposition 1  
 116 disappears: the lifted product differences out  $h$  automatically. Under the following generic-rank  
 117 condition, fewer nuisance dimensions cannot suffice for static linear-Gaussian sites:

118 **Proposition 3** (Lifting dimension). *Let  $x = ay + Du + e$  with mutually independent  $y \sim \mathcal{N}(0, 1)$ ,  
 119  $u \sim \mathcal{N}(0, I_K)$  and Gaussian errors  $e_j$  of positive variance,  $D \in \mathbb{R}^{P \times K}$ , and every  $a_j \neq 0$ . Suppose  
 120 that after deleting any row,  $L = [a, D]$  still contains two disjoint nonsingular  $(1 + K) \times (1 + K)$   
 121 submatrices, which holds for almost every  $L$  once  $P \geq 2K + 3$ . If a product of mask-independent  
 122 sites of the form (2) with  $o_j = x_j - c_j$ , fixed  $w_j, c_j$  and fixed  $\psi_j > 0$  (all independent of readings  
 123 and subsets), over  $z = (y, h)$  with  $h \in \mathbb{R}^{K'}$  and a proper nonsingular Gaussian prior, reproduces  
 124  $p(y | x_S)$  for every reading and every  $S$  with  $|S| \leq 2$ , then  $K' \geq K$ .*

125 The proof (Appendix A) shows that the conditionals on single sensors and pairs identify the covari-  
 126 ance of the sensors, which a product of such sites can only represent as a matrix of rank  $1 + K'$  plus  
 127 a diagonal; the identifiability argument of factor analysis [1] excludes this when  $K' < K$ . Our main  
 128 synthetic panels ( $K = 1, P = 5$ ) meet the condition, so for products of static sites one nuisance  
 129 dimension is necessary and sufficient.

130 **Closed-form initialization from the support.** Before any learning, we estimate the factor model  
 131 from  $\mathcal{D}$  alone. EM for a Gaussian with missing entries [4, 21] gives the joint mean and covariance  
 132 of  $(x, y)$  from the incomplete calibration rows. Then  $A = \text{Cov}(x, y)/\text{Var}(y)$ , and principal-factor  
 133 iterations on the residual covariance  $\text{Cov}(x) - AA^\top \text{Var}(y)$  give  $(D, \psi)$  [31, 33]. With sensors stan-  
 134 dardized by their support statistics,  $o_j = \tilde{x}_j + A_j \mu_y$ . We call the resulting predictor *FA-Gaussian*;  
 135 it is exact for every mask under the fitted model.

136 **Learned cavity refinement.** Real sensors are nonlinear, real noise is not exactly factor-structured,  
 137 and  $n = 48$  calibration rows make plug-in estimates noisy. We therefore refine the sites for  $R$   
 138 rounds. In each round, every observed sensor  $j$  removes its site from (3) to form the cavity  $q_{-j}(z) =$   
 139  $\mathcal{N}(m_{-j}, V_{-j})$ , the posterior over target and nuisance given the other sensors. The cavity predicts the  
 140 reading as  $\hat{x}_j = A_j(m_{-j,y} - \mu_y) + D_j^\top m_{-j,h}$ , with variance  $v_j = w_j^\top V_{-j} w_j + \psi_j$ . A shared two-  
 141 layer network  $f_\theta$  reads the reading  $\hat{x}_j$ , an encoding  $e_j$  of sensor  $j$ 's calibration scatter, the anchor  
 142  $(A_j, D_j, \log \psi_j)$ , the cavity  $(m_{-j}, \log V_{-j,yy}, V_{-j,yh})$ , the raw and standardized residuals of  $\tilde{x}_j$   
 143 against  $\hat{x}_j, \log v_j$  and the fraction of observed sensors. It outputs a correction  $(\delta_j^{\text{off}}, \delta_j^{\text{slope}}, \delta_j^D, s_j)$ ,  
 144 and the site is replaced by the local linearization of the sensor's response at the cavity mean  $\bar{y} =$   
 145  $m_{-j,y}$ :

$$\begin{aligned} w_j &= (A_j + \delta_j^{\text{slope}}, D_j + \delta_j^D), & o_j &= \tilde{x}_j - g_j(\bar{y}) + w_{j,y} \bar{y}, \\ g_j(\bar{y}) &= A_j(\bar{y} - \mu_y) + \delta_j^{\text{off}}, & \psi_j &\leftarrow \psi_j e^{4 \tanh(s_j/4)}. \end{aligned} \quad (4)$$

146 The encoding  $e_j \in \mathbb{R}^{16}$  is a DeepSets network [37] over sensor  $j$ 's calibration scatter, so  $f_\theta$  can  
 147 recognize a sensor's nonlinearity from the support. The last layer of  $f_\theta$  is initialized at zero.

148 If  $f_\theta$  were replaced by the statistical linear regression of a *known* response function under the cavity,  
 149 these rounds would be EP with posterior linearization [3, 7]. The network amortizes the unknown  
 150 response from the calibration rows, in the spirit of algorithm unrolling [11, 25].

151 **Proposition 4** (Structural guarantees). *For any parameters  $\theta$ :*

- 152 (i) *the prediction does not depend on the readings of missing sensors;*
- 153 (ii) *conditional on fully equivariant anchor parameters and support statistics with consistent*  
 154 *nuisance orientation under calibration-column permutations, permuting sensors and their*  
 155 *calibration columns leaves the prediction unchanged; the same  $\theta$  applies for every  $P$ ;*
- 156 (iii) *when the last layer of  $f_\theta$  is zero, the network equals the FA-Gaussian closed form of its*  
 157 *anchor, so the infimum of training risk over  $\theta$  is at most that closed form's.*

158 The original learned model does not enforce invariance to equivalent factor signs. Anchor refitting  
 159 can therefore violate end-to-end sensor-permutation invariance through a changed nuisance orienta-  
 160 tion. A separately versioned one-factor sign-averaging repair passed source-only structural checks;  
 161 the reported efficacy results remain those of the original model and do not establish the repaired  
 162 model's efficacy.

**Training.** We minimize the predictive NLL over episodes drawn from a pool of 60,000 synthetic tasks (128 tasks  $\times$  8 queries per step, 20,000 AdamW steps with cosine decay, query masks with at most one missing sensor). We use  $K = 1$  and  $R = 3$ . The model has about 8,000 parameters and trains in 15–20 minutes on one CPU core, 14 times less than the transformer baseline (Table 5). The same recipe trains every learned baseline except the transformer, which gets fresh tasks at every step.

## 5 Related work

**Products of experts** handle missing modalities by omission [13, 36, 22]; our negative result concerns scalar-target experts; MVAE already uses multivariate latent products. **Learned message passing**, through learned EP messages [12, 6, 16] or unrolled inference [11, 25], learns reusable operators for specified factors; here sensor response is inferred from deployment calibration support, with a target/nuisance cavity. **Supervised learning with missing values** [30, 21, 17] studies one task at a time; NeuMiss [20] unrolls the Bayes predictor of a Gaussian model but is trained per dataset and keyed to feature identities. **In-context prediction** with neural processes [8, 19] and prior-fitted networks such as TabPFN and TabICL [26, 14, 15, 28] uses generic set or transformer architectures (the cited TabICL is classification-only). Our tested transformer is trained on our task distribution; pretrained comparisons are pending. **Fusion under unknown correlations** [18, 35, 5, 29] includes learned sequential state-space filtering and decentralized network fusion. DIFNet explicitly discusses changing sensor nodes and network topologies. Our distinction is episodic prediction with unknown sensor calibration inferred from a labeled support set, rather than sequential filtering in a specified state-space model.

## 6 Experiments

### 6.1 Setup

**Synthetic tasks.** Each task draws, for every sensor  $j$ , a response  $x_j = a_j y + b_j + 0.4 \sin(f_j y) + d_j u + \sigma_j \epsilon_j$  with  $a_j = \pm \mathcal{U}[0.6, 1.8]$  (random sign),  $b_j \sim \mathcal{N}(0, 0.4^2)$ ,  $f_j \sim \mathcal{U}[0.5, 2.5]$ , shared-noise loading  $d_j \sim \mathcal{N}(0, 0.7^2)$ ,  $\sigma_j \sim \mathcal{U}[0.15, 0.8]$  and  $y, u, \epsilon_j \sim \mathcal{N}(0, 1)$ . The support has  $n = 48$  rows with 20% of sensor entries missing at random, and 48 query rows are scored under every mask with  $k$  missing sensors (or 20 seeded masks when there are more). The main panel (F1) has 256 fresh tasks with five sensors. Six further panels of 128 tasks change one factor at a time: 8 or 16 sensors, nonlinearity 0.8 or 0, and 16 or 96 support rows.

**Methods.** *Support-only closed forms:* FA-Gaussian, EM-Gaussian, NL-FA (cubic responses with a tuned curvature penalty and exact grid inference, a strong plug-in nonlinear model), evidence-tuned Bayesian linear regression (BLR), ridge, and, post hoc, a Gaussian process. *Learned in-context models*, all trained on the same task family and masks: our lifted cavity network (8k parameters, three seeds); a residual MLP that receives the same FA anchor and support encodings (15k parameters, five sensors only); the scalar-cavity network of §3; and a TabPFN-v2-style cell-token transformer [15] trained on fresh tasks at every step (203k parameters, 30,000 steps). A second transformer was trained on *all* masks, to separate architecture from training coverage. *Privileged references:* the oracle, and a sign-conditioned HMC approximation to the posterior predictive with the shared nuisance marginalized analytically (Appendix C).

**Protocol.** NLL is the primary metric; coverage, CRPS [10] and RMSE are secondary. Intervals are paired 95% intervals over tasks, with per-task NLL averaged over training seeds. Three evaluation protocols were committed to version control before their panels existed, and every endpoint is reported (Table 6, Appendix D).

### 6.2 Synthetic tasks with exact references

Table 1 and Figure 2a report the main panel. The lifted cavity network is the best non-privileged method for every number of missing sensors, including the two- and three-sensor deletions it never saw in training. With two missing sensors it closes 58% of the reference gap between FA-Gaussian

Table 1: Main synthetic panel (F1: 256 fresh tasks, five sensors). Mean test NLL by number of missing query sensors; learned models saw at most one missing sensor in training, so  $k = 2, 3$  are unseen patterns. Learned rows average three training seeds (seed s.d.  $\leq 0.002$ ) except the transformers (seed 1 as pre-registered; Table 11 adds a second seed). Last column: share of the reference gap (FA-Gaussian to sign-conditioned HMC) closed at  $k = 2$ . Best non-privileged entry per column in bold. <sup>†</sup>Post hoc, scored at  $k = 2$  only. <sup>‡</sup>Designed after protocol v2 (Section 6.6); descriptive on this panel and not bolded.

| Method                                                       | $k=0$         | $k=1$         | $k=2$        | $k=3$        | Gap closed |
|--------------------------------------------------------------|---------------|---------------|--------------|--------------|------------|
| <i>Privileged references</i>                                 |               |               |              |              |            |
| Oracle (true parameters)                                     | -0.419        | -0.250        | -0.026       | 0.287        | –          |
| Sign-conditioned HMC reference                               | -0.334        | -0.174        | 0.039        | 0.337        | –          |
| <i>Support-only closed forms</i>                             |               |               |              |              |            |
| FA-Gaussian                                                  | -0.152        | -0.011        | 0.177        | 0.439        | 0%         |
| EM-Gaussian                                                  | -0.048        | 0.037         | 0.195        | 0.441        | -13%       |
| NL-FA (cubic, tuned)                                         | -0.173        | -0.036        | 0.151        | 0.417        | 19%        |
| Bayesian linear regression                                   | 0.111         | 0.091         | 0.214        | 0.450        | -26%       |
| Ridge (complete case)                                        | 0.026         | 0.078         | 0.216        | 0.451        | -28%       |
| Ridge (mean-imputed)                                         | 0.198         | 0.280         | 0.399        | 0.582        | -161%      |
| GP, complete case <sup>†</sup>                               | –             | –             | 0.416        | –            | -173%      |
| <i>Learned in context (trained on <math>k \leq 1</math>)</i> |               |               |              |              |            |
| <b>Lifted cavity (ours), 8k</b>                              | <b>-0.265</b> | <b>-0.112</b> | <b>0.097</b> | <b>0.395</b> | 58%        |
| Transformer (TabPFN-v2 style), 203k                          | -0.246        | -0.084        | 0.131        | 0.424        | 34%        |
| Residual MLP on FA anchor, 15k                               | -0.236        | -0.087        | 0.114        | 0.397        | 46%        |
| Scalar-cavity network, 3k                                    | 0.055         | 0.178         | 0.336        | 0.555        | -115%      |
| Transformer, trained on all masks                            | -0.237        | -0.076        | 0.132        | 0.411        | 33%        |
| Lifted cavity transformer (v3) <sup>‡</sup> , 203k           | -0.265        | -0.112        | 0.095        | 0.387        | 59%        |

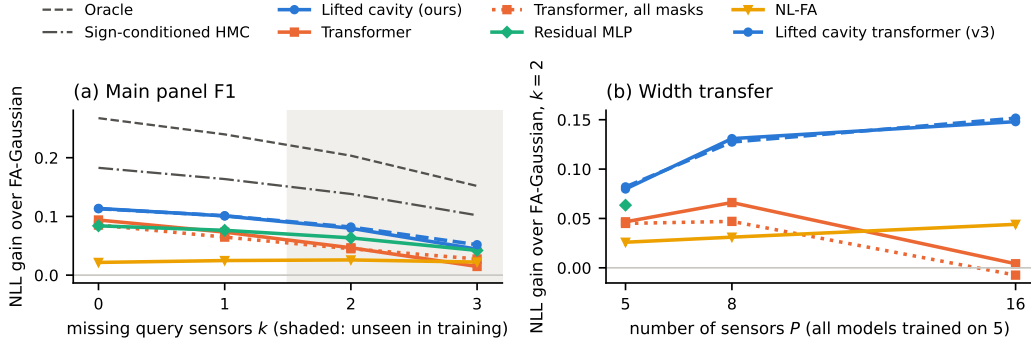


Figure 2: NLL gain over the closed-form FA-Gaussian anchor (higher is better). (a) Main panel F1 by number of missing query sensors; shaded patterns were never seen in training. Grey lines are the privileged oracle and the sign-conditioned HMC reference. (b) Width transfer with two missing sensors: every learned model was trained on five sensors, and the residual MLP exists only at  $P = 5$ . Dashed blue: the lifted cavity transformer of §6.6, descriptive on these panels.

and the sign-conditioned HMC approximation. Two comparators close less: the residual MLP, which receives the same anchor and support encodings with twice the parameters (46%), and the tuned nonlinear plug-in model NL-FA (19%). The TabPFN-v2-style transformer has 25 times the parameters and sees fresh tasks at every step, yet it is worse than the lifted cavity network for every number of missing sensors, by 0.020–0.033 nats (all intervals exclude zero), and closes 34% of the gap. Training it on every mask changes nothing (33%), so its deficit is learning efficiency, not unseen patterns. The residual MLP is the closest competitor: the lifted network beats it by 0.030 nats with all sensors observed and 0.016 with two missing, and the two tie with three missing. EM-Gaussian, BLR and both ridges fall below FA-Gaussian; mean imputation is the worst of all.

Table 2: Generalization beyond the training distribution (fresh panels of 128 tasks). Mean test NLL with two missing query sensors. Every learned model was trained on  $P=5$  sensors, nonlinearity 0.4 and  $n=48$  support rows; the residual MLP is defined only for five sensors. Best non-privileged entry per column in bold; italic rows are controls or descriptive (the lifted cavity transformer was designed after protocol v2) and are not bolded.

| Method                                | $P=8$         | $P=16$        | nonlin. 0.8  | linear       | $n=16$       | $n=96$       |
|---------------------------------------|---------------|---------------|--------------|--------------|--------------|--------------|
| Oracle                                | -0.598        | -1.142        | -0.086       | -0.066       | -0.060       | -0.043       |
| FA-Gaussian                           | -0.296        | -0.797        | 0.334        | <b>0.009</b> | 0.510        | 0.114        |
| EM-Gaussian                           | -0.160        | 2.977         | 0.311        | 0.034        | 1.140        | 0.112        |
| NL-FA                                 | -0.327        | -0.841        | 0.258        | 0.030        | 0.496        | 0.073        |
| BLR                                   | 0.615         | 0.058         | 0.331        | 0.061        | 3.690        | 0.125        |
| Lifted cavity (ours)                  | <b>-0.427</b> | <b>-0.945</b> | <b>0.191</b> | 0.031        | <b>0.363</b> | <b>0.052</b> |
| Transformer                           | -0.362        | -0.801        | <b>0.191</b> | 0.085        | 0.523        | 0.081        |
| Residual MLP                          | —             | —             | 0.235        | 0.036        | 0.371        | 0.074        |
| <i>Transformer, all masks</i>         | -0.343        | -0.790        | 0.190        | 0.080        | 0.483        | 0.080        |
| <i>Lifted cavity transformer (v3)</i> | -0.423        | -0.949        | 0.179        | 0.029        | 0.457        | 0.047        |

### 6.3 Generalization beyond the training distribution

**Width** (Table 2, Figure 2b). The lifted cavity network runs unchanged on 8 and 16 sensors, where it improves on FA-Gaussian by  $+0.131$  [ $+0.097$ ,  $+0.165$ ] and  $+0.148$  [ $+0.115$ ,  $+0.181$ ] nats with two missing sensors. The residual MLP cannot run on these panels, and EM-Gaussian degrades as the covariance grows. The transformer also runs at any width, but it gains less over FA-Gaussian as sensors are added. On 16 sensors it is no better than the closed form, and the lifted cavity network beats it by  $+0.144$  [ $+0.127$ ,  $+0.161$ ] nats. **Support size.** With 16 calibration rows the transformer degrades sharply, while the lifted cavity network keeps a  $+0.146$  [ $+0.073$ ,  $+0.219$ ] nat gain over FA-Gaussian; with 96 rows it remains the best method. **Stronger nonlinearity** (0.8 instead of 0.4) is the one shift where the transformer competes. It ties the lifted cavity network with two missing sensors (difference  $+0.000$  [ $-0.011$ ,  $+0.012$ ]) and is better by 0.041 nats when all sensors are observed: under strong response nonlinearity alone, Gaussian re-linearization offers no advantage over a generic transformer. **Linear sensors.** When the true sensors are linear, FA-Gaussian is nearly the right model and is best; the learned re-linearization then costs 0.022 nats with two missing sensors, and the transformer 0.076.

### 6.4 What the lifting buys

Ablations on F1 (Table 8, Appendix F) isolate the components. **Lifting** is the largest factor: scalar sites ( $K = 0$ ) with otherwise identical machinery and training lose 0.134 [ $+0.120$ ,  $+0.148$ ] nats with two missing sensors and end up worse than the untrained FA-Gaussian anchor. This supports lifting empirically under the tested training regime; Proposition 1 does not cover this context-conditioned ablation. **Cavity conditioning** adds 0.026 [ $+0.022$ ,  $+0.029$ ] nats on top of lifted static sites. A second nuisance dimension ( $K = 2$ ) changes nothing, as expected for one-factor noise. The scalar-cavity network of the earlier design, retrained with the same 60,000-task recipe, remains far behind FA-Gaussian (Appendix D).

### 6.5 Real sensor networks

**Beijing multi-site air quality [38].** The task is to predict log  $\text{PM}_{2.5}$  at each of 12 monitoring stations from the other 11, using hourly data with natural missingness. Each episode is one station and one week, with 48 labeled support hours and 48 query hours. The test period (2016-01 to 2017-02, 718 episodes) was never examined before protocol v2 was committed. Learned models are fine-tuned on episodes from 2013–2015 with one fixed recipe; the protocol’s analysis is cluster-robust over weeks.

With 11 sensors and 48 rows, the joint-covariance closed forms overfit, and evidence-tuned BLR is the strongest closed form. The fine-tuned lifted cavity network improves on it by 0.194 [ $+0.131$ ,  $+0.257$ ] nats under natural missingness, and by 0.137 [ $+0.104$ ,  $+0.170$ ] with six further sensors removed. The fine-tuned transformer is better still. The lifted cavity network’s gain over it is  $-0.118$

Table 3: Real sensor networks: mean test NLL on later, held-out periods. Beijing: 12 target stations  $\times$  weekly episodes in 2016–2017 with natural missingness, plus extra dropped sensors. Air Quality: weekly episodes after October 2004, sensor dropout simulated. Fine-tuning uses only earlier periods and one recipe for every model; lifted-cavity and residual-MLP rows average three seeds. Best entry per column in bold. <sup>†</sup>Post hoc. <sup>‡</sup>Designed after protocol v2 (Section 6.6); post hoc here and not bolded.

| Method                                                   | Beijing PM <sub>2.5</sub> (11 stations) |              |              | Air Quality CO |               | Air Quality NO <sub>2</sub> |               |
|----------------------------------------------------------|-----------------------------------------|--------------|--------------|----------------|---------------|-----------------------------|---------------|
|                                                          | natural                                 | +3           | +6           | $k=0$          | $k=2$         | $k=0$                       | $k=2$         |
| EM-Gaussian                                              | 1.032                                   | 0.742        | 0.569        | 0.074          | 0.085         | −0.237                      | −0.175        |
| FA-Gaussian                                              | 0.677                                   | 0.615        | 0.562        | 0.052          | 0.116         | −0.196                      | −0.155        |
| NL-FA                                                    | 0.938                                   | 0.744        | 0.610        | 0.144          | 0.091         | −0.090                      | −0.142        |
| Bayesian linear regression                               | 0.399                                   | 0.406        | 0.442        | 1.037          | 0.216         | −0.055                      | −0.178        |
| Ridge (complete case)                                    | 0.425                                   | 0.432        | 0.458        | 0.824          | 0.208         | −0.200                      | −0.176        |
| GP, complete case <sup>†</sup>                           | 1.884                                   | 1.312        | 0.934        | 3.364          | 0.503         | 25.217                      | −0.198        |
| $k$ -nearest neighbours <sup>†</sup>                     | 0.508                                   | 0.514        | 0.540        | 0.571          | 0.353         | −0.010                      | −0.063        |
| Lifted cavity, zero-shot                                 | 0.826                                   | 0.661        | 0.561        | 0.044          | 0.129         | −0.350                      | −0.233        |
| Transformer, zero-shot                                   | 0.958                                   | 0.743        | 0.687        | −0.017         | 0.081         | −0.360                      | −0.231        |
| <b>Lifted cavity, fine-tuned (ours)</b>                  | 0.205                                   | 0.237        | 0.305        | −0.040         | 0.008         | −0.412                      | −0.337        |
| no synthetic pretraining                                 | 0.247                                   | 0.279        | 0.344        | −0.001         | −0.006        | −0.419                      | −0.316        |
| no cavity input                                          | 0.259                                   | 0.286        | 0.349        | −0.086         | −0.028        | −0.423                      | −0.332        |
| scalar sites ( $K = 0$ )                                 | 0.215                                   | 0.252        | 0.347        | −0.026         | −0.003        | −0.359                      | −0.302        |
| Transformer, fine-tuned                                  | <b>0.087</b>                            | <b>0.121</b> | <b>0.236</b> | <b>−0.163</b>  | <b>−0.052</b> | <b>−0.485</b>               | <b>−0.381</b> |
| Residual MLP, fine-tuned                                 | —                                       | —            | —            | 0.025          | 0.110         | −0.246                      | −0.147        |
| <i>Lifted cavity transformer, fine-tuned<sup>‡</sup></i> | 0.122                                   | 0.160        | 0.259        | −0.099         | −0.027        | −0.404                      | −0.359        |

255 [−0.145, −0.091] nats under natural missingness and −0.069 [−0.080, −0.058] with six sensors  
 256 removed, so endpoint E4b fails. Before fine-tuning the order is reversed: zero-shot, the lifted cavity  
 257 network is ahead by 0.127 [+0.057, +0.197] nats with six sensors removed. Synthetic pretraining  
 258 helps: the same network fine-tuned from initialization is 0.041 nats worse [+0.033, +0.050].

259 **UCI Air Quality [2].** Five metal-oxide sensors on one device predict the reference analyzers’ log  
 260 CO and log NO<sub>2</sub>, in weekly episodes after October 2004. These weeks were examined in protocol  
 261 v1, so this is secondary evidence. With two simulated missing sensors, the fine-tuned lifted cavity  
 262 network improves on FA-Gaussian by +0.107 [+0.015, +0.200] nats for CO and +0.181 [+0.076,  
 263 +0.287] for NO<sub>2</sub>. The fine-tuned transformer is again better: the lifted network’s gain over it is  
 264 −0.060 [−0.106, −0.014] (CO) and −0.044 [−0.074, −0.014] (NO<sub>2</sub>). On CO, the cavity input  
 265 stops helping after fine-tuning.

266 **Why the transformer wins on real data.** The transformer is better on 89% of Beijing test  
 267 episodes, by 0.15–0.16 nats in each of the three easier quartiles of closed-form difficulty. The  
 268 lifted network underfits rather than overfits: the transformer’s final training NLL is 0.19 nats lower,  
 269 about the test gap. Neither a second nuisance dimension (0.205) nor five times more fine-tuning  
 270 steps (0.169, against 0.207; Table 10) closes the gap to the transformer (0.087), and per-episode  
 271 nonparametrics do worse still ( $k$ -nearest neighbours 0.508, Gaussian process 1.884). We therefore  
 272 hypothesized that the limitation lies in how sites are refined (each correction sees only its own sen-  
 273 sor’s reading, calibration scatter and cavity) rather than in the site family; protocol v3 tests this.

## 274 6.6 Lifted sites as a transformer output layer (protocol v3)

275 Protocol v2 left a split: structure wins on synthetic tasks, the transformer on real networks after fine-  
 276 tuning. The *lifted cavity transformer* (LCT) keeps the transformer’s backbone, recipe and seed and  
 277 replaces only its output head: a zero-initialized linear map turns each observed query cell’s final to-  
 278 ken into the correction ( $\delta_j^{\text{off}}, \delta_j^{\text{slope}}, \delta_j^D, s_j$ ) of that sensor’s site in (4), linearized at the FA-Gaussian  
 279 posterior mean, and the prediction is the  $y$ -marginal of the product of observed sites. Missing sen-  
 280 sors contribute no site and their readings never enter; at initialization the LCT is its anchor’s closed  
 281 form. Because the hypothesis was formed after the v2 results, v3 uses only new data, two fresh syn-  
 282 thetic panels (G1, five sensors; G3, sixteen) and two Beijing pollutants never scored before (NO<sub>2</sub>  
 283 and CO), and its four endpoints were committed before any of these existed (Table 6).

284 On the fresh synthetic panels, the lifted output layer helps the transformer (Table 4). With two miss-  
 285 ing sensors the LCT beats the transformer with the identical backbone, recipe and seed by 0.035

Table 4: Protocol v3 (new panels and targets). Mean test NLL. G1: 256 fresh five-sensor tasks by number of missing query sensors  $k$ ; G3: 128 fresh 16-sensor tasks,  $k = 2$ . Beijing: log NO<sub>2</sub> and log CO at each station from the other 11 stations, 2016–2017, natural missingness and six further sensors removed; learned models fine-tuned with the recipe of protocol v2. Best entry per column in bold.

| Method                                  | $k=0$         | G1, $P=5$     |              |              | G3<br>$P=16$  | Beijing NO <sub>2</sub> |              | Beijing CO   |              |
|-----------------------------------------|---------------|---------------|--------------|--------------|---------------|-------------------------|--------------|--------------|--------------|
|                                         |               | $k=1$         | $k=2$        | $k=3$        |               | natural                 | +6           | natural      | +6           |
| FA-Gaussian                             | −0.189        | −0.054        | 0.129        | 0.396        | −0.806        | 0.684                   | 0.506        | 0.384        | 0.334        |
| Bayesian linear regression              | 0.013         | 0.022         | 0.153        | 0.401        | 0.063         | 0.275                   | 0.365        | 0.170        | 0.248        |
| Lifted cavity network, 8k               | <b>−0.329</b> | <b>−0.172</b> | 0.040        | 0.348        | −0.934        | 0.282                   | 0.334        | 0.149        | 0.229        |
| Transformer, 203k                       | −0.315        | −0.147        | 0.073        | 0.378        | −0.800        | <b>0.199</b>            | 0.296        | <b>0.019</b> | <b>0.176</b> |
| <b>Lifted cavity transformer</b> , 203k | <b>−0.329</b> | <b>−0.172</b> | <b>0.039</b> | <b>0.340</b> | <b>−0.950</b> | 0.220                   | <b>0.291</b> | 0.054        | 0.187        |

[+0.027, +0.042] nats on G1 and 0.150 [+0.133, +0.167] on sixteen sensors (E5, E6 pass). It matches the small lifted network on five sensors (+0.002 [−0.002, +0.006]) and exceeds it on sixteen (+0.017 [+0.004, +0.029]), and it learns faster: its training NLL reaches the transformer’s final value after 10,750 of 30,000 steps (Figure 3). On the v2 panels (descriptive) it closes 59% of the F1 gap (lifted network 58%) but inherits part of the transformer’s degradation with 16 calibration rows. After fine-tuning on Beijing NO<sub>2</sub>, it beats the fine-tuned lifted network by +0.062 [+0.024, +0.100] nats (E7a passes), recovering 75% of the gap between the lifted network (0.282) and the transformer (0.199). Against the transformer the difference is −0.021 [−0.093, +0.052]; the interval includes zero but extends below the margin of −0.02, so non-inferiority (E7b) is not established. With six more stations missing, the two tie (+0.005 [−0.027, +0.037]). The real-data gain is seed-sensitive: with second seeds of both transformers (post hoc) the LCT recovers 50%, and 62% averaged over seeds (Table 11). The pattern repeats on Beijing CO (descriptive) and PM<sub>2.5</sub> (post hoc), with 73% and 70% of the gap recovered; on the small Air Quality device (post hoc) it is about half with two missing sensors (CO 58%, NO<sub>2</sub> 50%) and none for NO<sub>2</sub> with all sensors.

## 6.7 Pre-registered endpoints and calibration

Table 6 in Appendix D lists every declared endpoint, including pending v4 tests. Five of six v2 endpoints pass; the failure, E4b, is the real-data comparison with the fine-tuned transformer. All five v1 endpoints passed (Appendix D). Three of the four v3 endpoints pass; the failure, E7b, is non-inferiority to the fine-tuned transformer on Beijing NO<sub>2</sub>, where the interval is wide. Learned models are close to nominal 90% coverage (lifted cavity network 0.901 on F1, 0.914 on Beijing), while the closed forms under-cover (Table 9).

## 7 Limitations

**Scope.** Exact oracle and sign-conditioned HMC references exist only for our generator (one-factor noise, sinusoidal nonlinearity, Gaussian targets). Per-sensor dropout is simulated on Air Quality and mostly simulated on Beijing, whose natural missingness covers 1.6% of query readings; the gas-drift benchmark could not separate methods (Appendix D). **Model class.** Gaussian sites cannot represent multimodal posteriors; we used  $K = 1$  throughout. **Baselines.** The transformers were trained on a CPU budget (203k parameters, 30,000 steps); second seeds reproduce every synthetic comparison, but the LCT’s real-data gain varies with the seed. Pretrained TabPFN v2 and TabICL efficacy comparisons remain pending; GPU interface probes supply no comparative evidence.

## 8 Conclusion

Mask-independent Gaussian experts over the scalar target can mishandle shared noise. In the static linear-Gaussian class under the stated rank assumptions, target/nuisance sites represent factor noise exactly. A small lifted cavity network learns fusion more efficiently than the tested 25-times-larger transformer trained on the same tasks and extrapolates to unseen missingness patterns and wider sensor sets, but on real networks it underfits and a fine-tuned transformer is better. As the transformer’s output layer, lifted sites keep the extrapolation and recover much of that gap; closing the rest, richer site families and vector targets are natural next steps.

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## A Proofs

**Proposition 1.** Under the model,  $p(x_j | y) = \mathcal{N}(x_j; a_j y, c_j)$  with  $c_j = d_j^2 + \sigma_j^2$ .

If  $q(y | x_j) = p(y | x_j)$  for all  $x_j$ , then  $p(y) t_j(y; x_j) \propto p(y) p(x_j | y)$  as functions of  $y$ . Hence  $t_j(y; x_j) = \kappa_j(x_j) p(x_j | y)$ , which is Gaussian in  $y$  with precision  $a_j^2/c_j > 0$ . For a pair,  $q$  is therefore the naive-Bayes posterior  $p(y) p(x_j | y) p(x_k | y)$ . Its posterior mean is linear in  $(x_j, x_k)$  with coefficients

$$(a_j/c_j, a_k/c_k) / \pi_q, \quad \pi_q = 1 + a_j^2/c_j + a_k^2/c_k,$$

so the coefficient of  $x_j$  has the sign of  $a_j$ .

The Bayes posterior uses  $C = \text{Cov}(x_j, x_k | y)$ , with off-diagonal entry  $d_j d_k$ . Its mean coefficients are  $(C^{-1}a)^\top / \pi$ , with  $\pi = 1 + a^\top C^{-1}a$ . Both posteriors are Gaussian, so they coincide for all  $(x_j, x_k)$  if and only if  $C^{-1}a = \text{diag}(C)^{-1}a$ . Equivalently  $a = C \text{diag}(C)^{-1}a$ , whose off-diagonal components are  $d_j d_k a_k / c_k = 0$  and  $d_j d_k a_j / c_j = 0$ . With  $a_j a_k \neq 0$  this holds if and only if  $d_j d_k = 0$ .

For the sign flip, take  $a = (1, 1)$ ,  $d = (1, 0.5)$  and  $\sigma = (0.05, 0.05)$ . Then the Bayes coefficient of  $x_1$  is  $+0.499$  alone and  $-0.959$  in the pair.  $\square$

**Proposition 2.** Given  $z = (y, h)$ , the readings are independent:  $x_j | z \sim \mathcal{N}(w_j^\top z + c_j, \psi_j)$ . Hence

$$p(z | x_S) \propto p(z) \prod_{j \in S} \mathcal{N}(x_j; w_j^\top z + c_j, \psi_j).$$

In natural parameters each factor contributes  $w_j w_j^\top / \psi_j$  to the precision and  $w_j(x_j - c_j) / \psi_j$  to the shift, which is (2) with  $o_j = x_j - c_j$ . The prior is Gaussian, so the product is the Gaussian (3). This holds for every  $S$  because the factors of unobserved sensors are absent from both sides. Marginalizing  $h$  gives  $p(y | x_S)$ .  $\square$

**Proposition 3.** *Step 1: the product is a latent-factor model.* By the computation in the proof of Proposition 2 read backwards, a product of sites (2) with  $o_j = x_j - c_j$  is the exact posterior over  $z$  of the model  $M_q$ :  $z \sim \mathcal{N}(\mu_z, \Sigma_z)$  and, independently given  $z$ ,  $x_j \sim \mathcal{N}(w_j^\top z + c_j, \psi_j)$ . Its  $y$ -marginal is  $M_q$ 's conditional  $p_q(y | x_S)$ , and  $M_q$ 's sensor covariance is  $\Sigma_q = W \Sigma_z W^\top + \text{diag}(\psi)$  with  $W \in \mathbb{R}^{P \times (1+K')}$ , so  $\Sigma_q - \text{diag}(\psi)$  has rank at most  $1 + K'$ .

*Step 2: conditionals on  $|S| \leq 2$  identify the covariance.* Write  $V = \text{Var}(y)$ ,  $c = \text{Cov}(x, y)$  and  $\Sigma = \text{Cov}(x)$ . The case  $S = \emptyset$  gives  $V$ . For  $S = \{j\}$  the conditional has slope  $\beta_j = c_j / \Sigma_{jj}$  and variance  $r_j = V - c_j \beta_j$ , and  $\beta_j \neq 0$  because  $c_j = a_j \neq 0$ . Hence  $c_j = (V - r_j) / \beta_j$  and  $\Sigma_{jj} = c_j / \beta_j$ . For  $S = \{j, k\}$  the slope vector  $b \neq 0$  solves  $\Sigma_{\{jk\}} b = (c_j, c_k)$ , and whichever entry of  $b$  is non-zero determines  $\Sigma_{jk}$  from one row. Since  $M_q$  has the same conditionals, the same formulas apply to it, so  $\Sigma_q = \Sigma = LL^\top + \text{diag}(\sigma^2)$ .

*Step 3: fewer factors cannot reproduce  $\Sigma$ .* Suppose toward contradiction  $K' < K$ . Let  $r = 1 + K$ , so  $LL^\top$  has rank  $r$ . Suppose  $\Sigma = G + \Psi'$  with  $\Psi'$  diagonal and  $\text{rank}(G) \leq r$ . Fix a row  $i$  and two disjoint row sets  $I_1, I_2$  not containing  $i$  with  $L_{I_1}, L_{I_2}$  nonsingular. In the  $(r+1) \times (r+1)$  submatrix of  $\Sigma$  with rows  $\{i\} \cup I_1$  and columns  $\{i\} \cup I_2$ , every entry except  $(i, i)$  lies off the diagonal, so it is shared by  $G$  and  $LL^\top$ . Both  $G$  and  $LL^\top$  have rank at most  $r$ , so the corresponding minors vanish. Expanding along entry  $(i, i)$ , its cofactor is  $\det(L_{I_1} L_{I_2}^\top) \neq 0$ , so  $G_{ii} = (LL^\top)_{ii}$ . Hence  $G = LL^\top$ , which has rank  $r$  [1]. Applying this to  $G = W \Sigma_z W^\top$  from Step 1 gives  $1 + K' \geq r = 1 + K$ .

For almost every  $L$  all  $r \times r$  submatrices are nonsingular, and  $P \geq 2r + 1 = 2K + 3$  leaves two disjoint sets of  $r$  rows after deleting any row.  $\square$

**Proposition 4.**

- (i) Sites of missing sensors are multiplied by the mask before entering (3), and the network's per-sensor inputs never include another sensor's raw reading, only cavity statistics built from observed sites.
- (ii) Every operation is applied per sensor with shared weights. Conditional on a fully equivariant, consistently oriented anchor representation, its fitted sensor parameters permute with

the sensor coordinates, and sums over sensors are permutation-invariant. An unconstrained eigensolver can change factor signs or rotations when the anchor is refitted; the original learned network is not invariant to these changes. No input encodes a sensor’s index or the total number of sensors, except the fraction of observed sensors, which is permutation invariant.

(iii) With a zero last layer,  $\delta^{\text{off}} = \delta^{\text{slope}} = \delta^D = s = 0$ . Then  $w_j = (A_j, D_j)$ ,  $\psi_j$  is unchanged and  $o_j = \tilde{x}_j - A_j(\bar{y} - \mu_y) + A_j\bar{y} = \tilde{x}_j + A_j\mu_y$ . These are the initialization sites, so every round reproduces them and the output is the anchor’s FA-Gaussian closed form. The zero point lies in the parameter set, so the infimum of the training risk is at most that closed form’s.  $\square$

## B Architecture and training details

**Lifted cavity network ( $K = 1$ ,  $R = 3$ , 8,004 parameters).**

- **Support encoder.**  $\phi : \mathbb{R}^3 \rightarrow \mathbb{R}^{32} \rightarrow \mathbb{R}^{32}$  with tanh, applied to each observed calibration entry  $(\tilde{y}_i, \tilde{x}_{ij}, \tilde{r}_{ij})$ . The inputs are the standardized target, the centred reading, and the anchor residual scaled by  $\sqrt{\psi_j}$ . The encoder averages over observed rows and then applies  $\rho : \mathbb{R}^{33} \rightarrow \mathbb{R}^{16}$ , where the extra input is the observation rate.
- **Site network.**  $f_\theta : \mathbb{R}^{28} \rightarrow \mathbb{R}^{64} \rightarrow \mathbb{R}^{64} \rightarrow \mathbb{R}^4$  with tanh and a zero-initialized last layer.
- **Anchor.** EM uses 8 iterations with a  $10^{-4}$  ridge. Principal factors use 40 iterations, with uniquenesses floored at  $10^{-3}$ . Everything is computed in double precision from the support only.

Table 5: Compute. Training on one CPU core (core-hours); inference (ms/task) is single-threaded time per task for 480 queries (ten masks  $\times$  48 rows) on F1. Anchors are precomputed in batch and excluded; the closed-form rows include their unbatched NumPy fits.

| Model                         | Parameters | Training steps | Core-hours | ms/task |
|-------------------------------|------------|----------------|------------|---------|
| Lifted cavity network         | 8,004      | 20,000         | 0.22       | 7.2     |
| Residual MLP on FA anchor     | 15,490     | 20,000         | 0.20       | 1.2     |
| Transformer (TabPFN-v2 style) | 202,562    | 30,000         | 3.11       | 42.9    |
| Lifted cavity transformer     | 202,822    | 30,000         | 3.15       | 55.7    |
| FA-Gaussian (closed form)     | —          | —              | —          | 38.3    |
| NL-FA (closed form)           | —          | —              | —          | 34.4    |

**Training recipe (all synthetic-trained models except the transformer).**

- Pool of 60,000 tasks (seed 5150).
- Minibatches of 128 tasks  $\times$  8 queries.
- AdamW with learning rate  $1.8 \times 10^{-3}$ , cosine decay to 10% and weight decay  $10^{-4}$ .
- Gradient clipping at 5 and 20,000 steps.
- Seeds 1–3.

No hyperparameter search was run for any model.

**Controls.**

- **Residual MLP on the same anchor** (15,490 parameters). It receives the FA-Gaussian prediction, the same per-sensor support encodings (concatenated in sensor order), the readings and mask, and a 55-dimensional support summary. It outputs a residual correction to the FA-Gaussian mean and log-variance, and is defined only for five sensors.
- **Lifted sites without cavity input.** The same sites with the cavity statistics removed.
- **1-D sites** ( $K = 0$ ). The same machinery, with sites over  $y$  only.
- **Repository cavity network.** The three-round scalar-cavity architecture of Appendix D, retrained with the shared recipe.

511 **Transformer baseline (202,562 parameters).** The transformer follows TabPFN v2 [15]:

- 512 • every cell is a token, embedded by a linear map of its value or by a learned missing-value token;
- 513
- 514 • sensors receive random per-task feature embeddings, so the model is width-agnostic and
- 515 permutation equivariant in distribution;
- 516 • the target column has its own embedding, and query targets use a learned token;
- 517 • each of 4 blocks alternates attention across the columns of a row with attention across rows
- 518 within a column (all rows attend to the support rows only), followed by an MLP ( $d = 64$ ,
- 519 4 heads, hidden width 128, pre-LayerNorm);
- 520 • targets are standardized by the support mean and standard deviation, and the output is a
- 521 Gaussian mean and log-variance.

522 It was trained on fresh tasks at every step (16 tasks  $\times$  16 queries, AdamW with learning rate  $10^{-3}$ ,  
523 1,000 warm-up steps, cosine decay, clipping at 1, 30,000 steps), with the same query masks as  
524 the other models. A second copy (*all masks*) saw query masks with 0–4 missing sensors, drawn  
525 uniformly. Evaluation uses a fixed seed for the random feature embeddings.

526 **Lifted cavity transformer (protocol v3, 202,822 parameters).** The backbone is the transformer  
527 above, unchanged. Its output head is replaced by a zero-initialized linear map from each query cell’s  
528 final token to four numbers ( $\delta_j^{\text{off}}, \delta_j^{\text{slope}}, \delta_j^D, s_j$ ), used exactly as in (4) with the linearization point  $\bar{y}$   
529 set to the anchor’s closed-form posterior mean (one round,  $K = 1$ , gradients not propagated through  
530  $\bar{y}$ ). Readings of missing sensors are replaced by the missing-value token before the backbone and  
531 contribute no site, so they never enter the prediction (a unit test checks both this and the identity  
532 with the anchor’s closed form at initialization). Anchors are computed on the fly for each training  
533 task. Training, masks, seed and evaluation are those of the transformer. Figure 3 compares the two  
534 training curves.

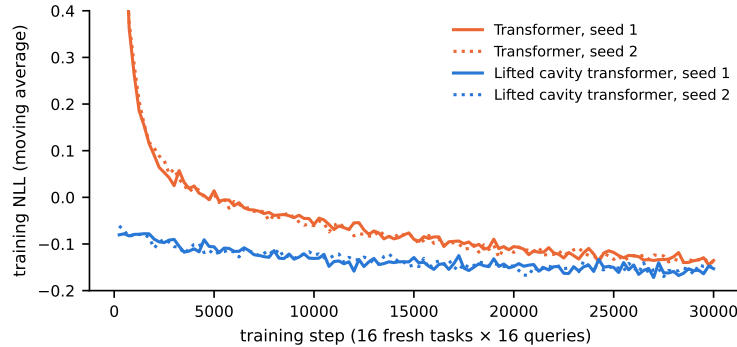


Figure 3: Training NLL (moving average; fresh tasks with at most one missing sensor) of the transformer and the lifted cavity transformer, which share backbone, recipe and seed. The lifted output layer starts at its anchor’s closed form, so it begins far lower; the plain transformer must first learn fusion from scratch.

## 535 C Baselines and references

### 536 Support-only closed forms.

- 537 • **EM-Gaussian:** the conditional of  $y$  given  $x_S$  under the EM joint Gaussian (40 iterations).
- 538 • **FA-Gaussian:** the closed form of §4, with EM run for 40 iterations. The networks’ anchors
- 539 use 8 iterations and no variance inflation, so the reported FA-Gaussian is a slightly stronger
- 540 baseline than the untrained networks (on F1 with two missing sensors, 0.177 against 0.187).
- 541 • **NL-FA:** per-sensor cubic response curves in the standardized target. The intercept and
- 542 slope are unpenalized; the curvature terms are ridge-penalized ( $\lambda = 100$ ) and clamped to
- 543 the support’s target range for linear extrapolation. The residual covariance has a one-factor

Table 6: Every declared endpoint of protocols v2–v4. Protocols v2/v3 preceded their panels; v4 was amended after partial data/reference inspection, before compared-model scoring (protocol v1: Appendix D). Gain in nats of NLL, positive when the first-named model is better; paired 95% intervals, cluster-robust over weeks on Beijing. v4 endpoints average three training seeds per task. Pass: gain  $\geq 0.01$  and lower bound  $> 0$ , except the non-inferiority endpoints E7b and E10 (lower bound  $> -0.02$ ).

| ID  | Data                      | Comparison                                        | Gain [95% CI]           | Result      |
|-----|---------------------------|---------------------------------------------------|-------------------------|-------------|
| E1  | F1, $k = 2$               | lifted cavity network vs NL-FA                    | +0.054 [+0.043, +0.065] | pass        |
| E2  | F1, $k = 2$               | lifted cavity network vs transformer              | +0.033 [+0.027, +0.040] | pass        |
| E3a | F3 (16 sensors), $k = 2$  | lifted cavity network vs FA-Gaussian              | +0.148 [+0.115, +0.181] | pass        |
| E3b | F3 (16 sensors), $k = 2$  | lifted cavity network vs transformer              | +0.144 [+0.127, +0.161] | pass        |
| E4a | Beijing PM <sub>2.5</sub> | lifted cavity network (FT) vs BLR                 | +0.194 [+0.131, +0.257] | pass        |
| E4b | Beijing PM <sub>2.5</sub> | lifted cavity network (FT) vs transformer (FT)    | −0.118 [−0.145, −0.091] | <b>fail</b> |
| E5  | G1, $k = 2$               | LCT vs transformer                                | +0.035 [+0.027, +0.042] | pass        |
| E6  | G3 (16 sensors), $k = 2$  | LCT vs transformer                                | +0.150 [+0.133, +0.167] | pass        |
| E7a | Beijing NO <sub>2</sub>   | LCT (FT) vs lifted cavity network (FT)            | +0.062 [+0.024, +0.100] | pass        |
| E7b | Beijing NO <sub>2</sub>   | LCT (FT) vs transformer (FT), non-inferiority     | −0.021 [−0.093, +0.052] | <b>fail</b> |
| E8  | H1, $k = 2$               | LCT-L vs transformer-L                            | pending                 | –           |
| E9  | H3 (16 sensors), $k = 2$  | LCT-L vs transformer-L                            | pending                 | –           |
| E10 | Beijing-new               | LCT-L (FT) vs transformer-L (FT), non-inferiority | pending                 | –           |
| E11 | Beijing-new, +6 removed   | LCT-L (FT) vs transformer-L (FT)                  | pending                 | –           |
| E12 | Beijing-new               | LCT-L (FT) vs TabPFN v2 (in context)              | pending                 | –           |

structure. Exact inference is done on a 401-point grid. The degree and  $\lambda$  were tuned on the development panel only: free degree selection by leave-one-out produced catastrophic extrapolation.

- **BLR**: evidence-maximized Bayesian linear regression [23] on complete-case rows, with predictive variance including parameter uncertainty.
- **Ridge**: complete-case ridge with leave-one-out residual variance, and the original mean-imputed ridge.
- **Gaussian process (post hoc)**: for each episode and observed-sensor set, an ARD squared-exponential kernel plus white noise on the complete-case calibration rows, with hyperparameters by type-II maximum likelihood (one restart).
- **$k$ -nearest neighbours (post hoc)**: on the standardized observed sensors of complete-case calibration rows, with  $k \in \{1, 2, 3, 5, 7, 10\}$  chosen per episode and sensor set by leave-one-out NLL on the support and the leave-one-out mean squared error as predictive variance.

EM-Gaussian, FA-Gaussian and ridge inflate their predictive variance by  $n/(n - |S| - 1)$ .

**Sign-conditioned HMC reference.** The reference uses HMC [27] over the 25 (or 5 $P$ ) task parameters. It uses the generator priors, transformed to unconstrained coordinates. The shared nuisance is marginalized analytically through a rank-one-plus-diagonal covariance, using only the observed entries of each calibration row. The sign of each  $a_j$  is fixed by the support correlation. The lower bound  $|a_j| \geq 0.6$  does not guarantee correct sign identification from a finite noisy support. Consequently this reference omits posterior uncertainty over response signs and is not a certified Bayes-optimal ceiling.

The sampler runs 4 chains per task, each started from a MAP fit. Warm-up has two phases: step-size adaptation (target acceptance 0.75) and diagonal mass-matrix estimation from 150 iterations, followed by 150 more adaptation iterations. We keep 50 draws per chain, thinned by 4, with 12 leapfrog steps.

The predictive density of a query uses the Monte Carlo samples, including the query’s own reweighting:  $p(y | x_S, \mathcal{D}) = \sum_m p(y, x_S | \theta_m) / \sum_m p(x_S | \theta_m)$ , with  $p(x_S | \theta_m)$  by trapezoid quadrature on 301 points. On every panel where it was run, R-hat on identifiable parameters was below 1.1 for at least 99.9% of parameters. Run with the true parameters, the predictive formula reproduces the exact oracle to within  $2 \times 10^{-3}$  nats (a unit test).

## D Protocols and audit trail

**How the method arose.** The lifted cavity network replaced an earlier design that failed. That design was a three-round EP-style network whose sites live on  $y$  alone and are updated from the scalar cavity of  $y$ . It was evaluated under pre-registered gates on the same synthetic family. Its first version collapsed to the prior during training. A repaired version, with sites initialized from support moments, trained successfully but was beaten by a ridge regression. On that panel the oracle, the ridge control and the best repaired seed scored:

| Method                                     | NLL, two sensors missing |
|--------------------------------------------|--------------------------|
| Oracle                                     | −0.046                   |
| Ridge control                              | 0.396                    |
| Repaired scalar-cavity network (best seed) | 0.403                    |

Computing the oracle exposed three problems. First, the control was weak: it mean-imputed missing calibration entries, and a support-only EM or factor-analysis Gaussian scores 0.168 or 0.156. Second, scalar sites are structurally limited (§3). Third, the learners were starved: 512 reused tasks and 3,000 updates. Retraining the scalar-cavity architecture with our recipe (60,000 tasks, 20,000 steps) improves it by only about 0.05 nats, and on F1 it remains 0.159 nats worse than the untrained FA-Gaussian (§6.4).

**Protocol v1.** After development on panels that are now retired, five endpoints were committed to version control before their panels were generated:

- fresh synthetic tasks, against FA-Gaussian and against the residual MLP;
- 8-sensor transfer;
- stronger nonlinearity;
- a new real target (NO<sub>2</sub> on the Air Quality device).

All five passed (Table 7). Each endpoint was evaluated once, and no model was retrained after the commit.

Table 7: Protocol v1 endpoints (two missing sensors; positive gain favours the lifted cavity network; pass requires gain  $\geq 0.01$  and a positive lower bound).

| Endpoint | Comparison                                                   | Gain [95% CI]           | Result |
|----------|--------------------------------------------------------------|-------------------------|--------|
| E1       | Fresh tasks, vs FA-Gaussian                                  | +0.074 [+0.064, +0.084] | pass   |
| E2       | Fresh tasks, vs residual MLP                                 | +0.016 [+0.012, +0.020] | pass   |
| E3       | 8 sensors, vs FA-Gaussian                                    | +0.115 [+0.097, +0.132] | pass   |
| E4       | Nonlinearity 0.8, vs residual MLP                            | +0.028 [+0.020, +0.036] | pass   |
| R1       | Air Quality NO <sub>2</sub> , 23 later weeks, vs EM-Gaussian | +0.112 [+0.050, +0.173] | pass   |

**Protocol v2.** Protocol v2 was likewise committed before any of its panels or test splits existed. It adds:

- the transformer baseline, NL-FA, BLR and the sign-conditioned HMC reference;
- wider panels (8 and 16 sensors), and shifts in nonlinearity and support size;
- the Beijing benchmark, whose test period had never been scored.

NL-FA’s settings and BLR’s choice as the strongest real-data closed form were fixed on development data (Appendix C). The gas-sensor-drift dataset [34] was examined during development and then excluded, before any test split was built. Its heavy-tailed sensor outliers made the mean NLL of every Gaussian method dominated by a few episodes, with mean and median differing by more than 100 nats, so it cannot separate methods. One deviation was recorded before any Beijing score existed: some query hours have no other station observed, and complete-case BLR cannot fit zero features, so for an empty sensor set it returns the support mean with variance  $s^2(1 + 1/n)$ , like the other closed forms’ prior predictive.

**Protocol v3.** After protocol v2’s real-data endpoint failed, protocol v3 was committed before any of its panels or targets existed. It fixes one model (the lifted cavity transformer of §6.6, trained once with the transformer’s recipe and seed and evaluated at its final checkpoint), two new synthetic panels (G1, G3), two Beijing pollutants never scored by any model (NO<sub>2</sub>, primary; CO, descriptive), the fine-tuning recipe of protocol v2, and four endpoints (Table 6). The lifted cavity transformer’s scores on the v2 panels are descriptive only (Appendix F). Two container restarts interrupted its training before any v3 score existed; it was retrained from scratch with the same seed, and later runs checkpoint every random state so that a resumed run is bitwise identical to an uninterrupted one (a unit test). The protocol document logs both events.

**Protocol v4 status.** The larger three-seed comparison and pretrained TabPFN evaluation are still in progress. No v4 efficacy endpoint is reported in this version.

**Protocol v4.** Protocol v4 was committed after protocol v3’s results and before any of its panels or targets existed. It fixes:

- two models, each trained on a GPU with three seeds: transformer-L and LCT-L, with identical 2.1M-parameter backbones and fixed 40,000-step recipes;
- the pretrained TabPFN v2 regressor, used in context only, with eight estimators and a fixed random state;
- two fresh synthetic panels (H1, H3);
- three Beijing pollutants never scored by any model (PM<sub>10</sub>, SO<sub>2</sub> and O<sub>3</sub>, pooled as Beijing-new);
- the fine-tuning recipe of protocol v2;
- five endpoints (Table 6).

**Amendment.** One amendment was recorded before any compared model was trained or scored on a new target. Building the Beijing-new panels showed two things:

- The readings are integers, and 34.2% of SO<sub>2</sub> and 12.8% of O<sub>3</sub> test readings equal the floor value  $2 \mu\text{g m}^{-3}$ .
- Three SO<sub>2</sub> support sets were constant.

A continuous NLL can reward concentrating density on quantization values. Histogram and Gaussian outputs can benefit differently from this, complicating their comparison. We therefore dequantize [32]. Every integer reading of every station becomes reading  $+\mathcal{U}(-\frac{1}{2}, \frac{1}{2})$  before the log. The noise is fixed per pollutant and shared by the test and fine-tuning episodes. Readings on finer grids, which occur only in 2013–2015, are kept as they are. Before the amendment, the only outputs seen were the closed-form reference scores printed while the undequantized panels were built; those panels were discarded.

**GPU-side changes.** Every change on the GPU side affects speed or robustness only:

- batch prefetching;
- longer checkpoint intervals, with a fallback to the previous checkpoint;
- a cached fine-tuning pool.

An interrupted 12-update LCT CPU run resumes to bitwise-identical weights in the unit test; this does not establish bitwise reproducibility of the full GPU runs or every execution optimization. TabPFN predicts all conditions of an episode in one call; its query rows do not interact. The third training seeds of the 203k transformers are post hoc.

**Interpretation limits.** The five decisions use nominal endpoint-wise 95% intervals, without a multiplicity correction or global superiority rule. Averaging three fitted seeds within each task conditions on those fits; the intervals omit training-randomness uncertainty. Week clustering does not establish independence between adjacent weeks. E12 compares target-source-fine-tuned LCT-L with in-context-only TabPFN: episode support is matched, but prior target-specific training is unequal, so this comparison cannot isolate the architecture. The real-data estimand is contemporaneous within-week adaptation, not forecasting.

Table 8: Ablations on F1 (mean test NLL). Same training recipe for every learned row.

| Variant                                  | $k=0$  | $k=1$  | $k=2$ | $k=3$ |
|------------------------------------------|--------|--------|-------|-------|
| Lifted cavity, $K = 1$ (ours)            | -0.265 | -0.112 | 0.097 | 0.395 |
| Lifted cavity, $K = 2$                   | -0.267 | -0.112 | 0.098 | 0.397 |
| Lifted sites, no cavity input            | -0.221 | -0.074 | 0.123 | 0.399 |
| Scalar sites ( $K = 0$ ), same machinery | -0.070 | 0.061  | 0.232 | 0.473 |
| Scalar-cavity network (retrained)        | 0.055  | 0.178  | 0.336 | 0.555 |
| FA-Gaussian (untrained)                  | -0.152 | -0.011 | 0.177 | 0.439 |

Table 9: Calibration and point accuracy. Coverage of the central 90% predictive interval (target 0.90), continuous ranked probability score (CRPS, lower is better) and RMSE. F1 with two missing sensors; Beijing test with natural missingness (fine-tuned learned models).

| Method               | F1, $k = 2$ |       |       | Beijing, natural |       |       |
|----------------------|-------------|-------|-------|------------------|-------|-------|
|                      | Cov. 90     | CRPS  | RMSE  | Cov. 90          | CRPS  | RMSE  |
| FA-Gaussian          | 0.873       | 0.168 | 0.316 | 0.824            | 0.189 | 0.379 |
| EM-Gaussian          | 0.860       | 0.170 | 0.318 | 0.807            | 0.207 | 0.423 |
| BLR                  | 0.866       | 0.174 | 0.327 | 0.861            | 0.182 | 0.374 |
| Lifted cavity (ours) | 0.901       | 0.160 | 0.305 | 0.914            | 0.171 | 0.357 |
| Transformer          | 0.902       | 0.165 | 0.314 | 0.914            | 0.158 | 0.341 |
| Residual MLP         | 0.902       | 0.162 | 0.309 | —                | —     | —     |

**Post-hoc analyses.** Everything not named in a protocol is post hoc and labeled as such: the per-episode Gaussian process and  $k$ -nearest neighbours, the second training seeds of both transformers, the fine-tuning diagnostics of §6.5, and the exploratory runs of Appendix F.

## E Broader impact

Calibrating low-cost environmental sensors widens access to air-quality monitoring. Overconfident predictions could mislead exposure decisions, which is why we report coverage alongside likelihood. We recommend validating calibration on each new deployment before relying on predicted intervals.

## F Additional results

Table 10: Post-hoc fine-tuning variants on Beijing  $\text{PM}_{2.5}$  (test period of protocol v2; mean test NLL). Neither more nuisance dimensions nor longer fine-tuning closes the gap to the fine-tuned transformer.

| Fine-tuned model                                | natural | +3    | +6    |
|-------------------------------------------------|---------|-------|-------|
| Lifted cavity network, standard recipe (seed 1) | 0.207   | 0.237 | 0.304 |
| with $K = 2$                                    | 0.205   | 0.239 | 0.300 |
| with $5 \times$ fine-tuning steps               | 0.169   | 0.201 | 0.276 |
| Transformer, standard recipe                    | 0.087   | 0.121 | 0.236 |

Table 11: Second training seeds of the transformer and the lifted cavity transformer (post hoc). Endpoint comparisons recomputed with each seed and with both averaged; the lifted cavity network always averages its three seeds. Gains in nats, paired 95% intervals (cluster-robust over weeks on Beijing).

| ID  | Data                      | Comparison                 | Seed 1 (protocol)       | Seed 2                  | Seeds averaged          |
|-----|---------------------------|----------------------------|-------------------------|-------------------------|-------------------------|
| E2  | F1, $k = 2$               | lifted vs transformer      | +0.033 [+0.027, +0.040] | +0.037 [+0.030, +0.044] | +0.035 [+0.029, +0.042] |
| E3b | F3, $k = 2$               | lifted vs transformer      | +0.144 [+0.127, +0.161] | +0.155 [+0.139, +0.171] | +0.149 [+0.134, +0.165] |
| E5  | G1, $k = 2$               | LCT vs transformer         | +0.035 [+0.027, +0.042] | +0.037 [+0.028, +0.046] | +0.036 [+0.028, +0.044] |
| E6  | G3, $k = 2$               | LCT vs transformer         | +0.150 [+0.133, +0.167] | +0.161 [+0.142, +0.180] | +0.155 [+0.139, +0.172] |
| E4b | Beijing PM <sub>2.5</sub> | lifted vs transformer (FT) | -0.118 [-0.145, -0.091] | -0.115 [-0.159, -0.070] | -0.116 [-0.150, -0.083] |
| E7a | Beijing NO <sub>2</sub>   | LCT vs lifted (FT)         | +0.062 [+0.024, +0.100] | +0.042 [-0.023, +0.108] | +0.052 [+0.002, +0.103] |
| E7b | Beijing NO <sub>2</sub>   | LCT vs transformer (FT)    | -0.021 [-0.093, +0.052] | -0.042 [-0.097, +0.012] | -0.032 [-0.093, +0.030] |

Table 12: Panel F2 (8 sensors; 128 tasks): mean test NLL by number of missing sensors.

| Method                         | $k=0$  | $k=1$  | $k=2$  | $k=3$  |
|--------------------------------|--------|--------|--------|--------|
| Oracle                         | -0.807 | -0.712 | -0.598 | -0.459 |
| FA-Gaussian                    | -0.462 | -0.387 | -0.296 | -0.183 |
| EM-Gaussian                    | 0.032  | -0.133 | -0.160 | -0.120 |
| NL-FA                          | -0.485 | -0.414 | -0.327 | -0.215 |
| BLR                            | 0.187  | 0.398  | 0.615  | 0.028  |
| Ridge (complete case)          | 0.064  | -0.003 | -0.033 | -0.015 |
| Lifted cavity (ours)           | -0.615 | -0.530 | -0.427 | -0.302 |
| Lifted cavity, $K = 2$         | -0.618 | -0.531 | -0.427 | -0.302 |
| Lifted sites, no cavity        | -0.551 | -0.469 | -0.369 | -0.250 |
| Scalar sites ( $K = 0$ )       | -0.295 | -0.228 | -0.148 | -0.057 |
| Transformer                    | -0.568 | -0.474 | -0.362 | -0.229 |
| Transformer, all masks         | -0.545 | -0.452 | -0.343 | -0.212 |
| Lifted cavity transformer (v3) | -0.611 | -0.526 | -0.423 | -0.299 |

Table 13: Panel F3 (16 sensors; 128 tasks): mean test NLL by number of missing sensors.

| Method                         | $k=0$  | $k=1$  | $k=2$  | $k=3$  |
|--------------------------------|--------|--------|--------|--------|
| Oracle                         | -1.223 | -1.184 | -1.142 | -1.096 |
| FA-Gaussian                    | -0.864 | -0.832 | -0.797 | -0.758 |
| EM-Gaussian                    | 6.300  | 4.851  | 2.977  | 1.553  |
| NL-FA                          | -0.910 | -0.876 | -0.841 | -0.803 |
| BLR                            | 0.095  | 0.078  | 0.058  | 0.054  |
| Ridge (complete case)          | 0.093  | 0.089  | 0.081  | 0.086  |
| Lifted cavity (ours)           | -1.018 | -0.982 | -0.945 | -0.902 |
| Lifted cavity, $K = 2$         | -1.017 | -0.981 | -0.944 | -0.901 |
| Lifted sites, no cavity        | -0.952 | -0.916 | -0.880 | -0.836 |
| Scalar sites ( $K = 0$ )       | -0.658 | -0.627 | -0.591 | -0.554 |
| Transformer                    | -0.886 | -0.846 | -0.801 | -0.747 |
| Transformer, all masks         | -0.869 | -0.832 | -0.790 | -0.738 |
| Lifted cavity transformer (v3) | -1.020 | -0.985 | -0.949 | -0.905 |

Table 14: Panel F4 (nonlinearity 0.8; 128 tasks): mean test NLL by number of missing sensors.

| Method                         | $k=0$  | $k=1$  | $k=2$  | $k=3$ |
|--------------------------------|--------|--------|--------|-------|
| Oracle                         | -0.513 | -0.331 | -0.086 | 0.261 |
| FA-Gaussian                    | 0.050  | 0.170  | 0.334  | 0.568 |
| EM-Gaussian                    | 0.050  | 0.145  | 0.311  | 0.559 |
| NL-FA                          | -0.032 | 0.089  | 0.258  | 0.507 |
| BLR                            | 0.208  | 0.188  | 0.331  | 0.568 |
| Ridge (complete case)          | 0.138  | 0.208  | 0.338  | 0.571 |
| Lifted cavity (ours)           | -0.185 | -0.026 | 0.191  | 0.498 |
| Lifted cavity, $K = 2$         | -0.200 | -0.035 | 0.186  | 0.497 |
| Lifted sites, no cavity        | -0.068 | 0.075  | 0.261  | 0.519 |
| Scalar sites ( $K = 0$ )       | -0.013 | 0.130  | 0.314  | 0.570 |
| Transformer                    | -0.226 | -0.043 | 0.191  | 0.503 |
| Transformer, all masks         | -0.221 | -0.038 | 0.190  | 0.486 |
| Residual MLP                   | -0.107 | 0.039  | 0.235  | 0.508 |
| Scalar-cavity network          | 0.133  | 0.256  | 0.416  | 0.635 |
| Lifted cavity transformer (v3) | -0.196 | -0.037 | 0.179  | 0.482 |

Table 15: Panel F5 (linear sensors; 128 tasks): mean test NLL by number of missing sensors.

| Method                         | $k=0$  | $k=1$  | $k=2$  | $k=3$ |
|--------------------------------|--------|--------|--------|-------|
| Oracle                         | -0.455 | -0.288 | -0.066 | 0.244 |
| FA-Gaussian                    | -0.362 | -0.203 | 0.009  | 0.305 |
| EM-Gaussian                    | -0.244 | -0.142 | 0.034  | 0.308 |
| NL-FA                          | -0.337 | -0.178 | 0.030  | 0.318 |
| BLR                            | -0.135 | -0.085 | 0.061  | 0.322 |
| Ridge (complete case)          | -0.169 | -0.092 | 0.062  | 0.323 |
| Lifted cavity (ours)           | -0.323 | -0.172 | 0.031  | 0.318 |
| Lifted cavity, $K = 2$         | -0.320 | -0.170 | 0.033  | 0.318 |
| Lifted sites, no cavity        | -0.343 | -0.189 | 0.018  | 0.309 |
| Scalar sites ( $K = 0$ )       | -0.085 | 0.031  | 0.186  | 0.415 |
| Transformer                    | -0.278 | -0.122 | 0.085  | 0.371 |
| Transformer, all masks         | -0.281 | -0.123 | 0.080  | 0.353 |
| Residual MLP                   | -0.304 | -0.161 | 0.036  | 0.319 |
| Scalar-cavity network          | 0.052  | 0.161  | 0.302  | 0.505 |
| Lifted cavity transformer (v3) | -0.319 | -0.171 | 0.029  | 0.315 |

Table 16: Panel F6 (16 support rows; 128 tasks): mean test NLL by number of missing sensors.

| Method                         | $k=0$  | $k=1$  | $k=2$  | $k=3$ |
|--------------------------------|--------|--------|--------|-------|
| Oracle                         | -0.446 | -0.280 | -0.060 | 0.252 |
| FA-Gaussian                    | 0.256  | 0.371  | 0.510  | 0.686 |
| EM-Gaussian                    | 24.900 | 4.682  | 1.140  | 0.739 |
| NL-FA                          | 0.314  | 0.388  | 0.496  | 0.661 |
| BLR                            | 0.898  | 0.857  | 3.690  | 0.887 |
| Ridge (complete case)          | 0.317  | 0.391  | 0.500  | 0.624 |
| Lifted cavity (ours)           | 0.073  | 0.195  | 0.363  | 0.604 |
| Lifted cavity, $K = 2$         | 0.156  | 0.250  | 0.392  | 0.614 |
| Lifted sites, no cavity        | 0.117  | 0.234  | 0.389  | 0.604 |
| Scalar sites ( $K = 0$ )       | 0.204  | 0.293  | 0.428  | 0.644 |
| Transformer                    | 0.253  | 0.371  | 0.523  | 0.731 |
| Transformer, all masks         | 0.256  | 0.359  | 0.483  | 0.653 |
| Residual MLP                   | 0.094  | 0.211  | 0.371  | 0.594 |
| Scalar-cavity network          | 0.345  | 0.439  | 0.565  | 0.750 |
| Lifted cavity transformer (v3) | 0.237  | 0.333  | 0.457  | 0.643 |

Table 17: Panel F7 (96 support rows; 128 tasks): mean test NLL by number of missing sensors.

| Method                         | $k=0$  | $k=1$  | $k=2$  | $k=3$ |
|--------------------------------|--------|--------|--------|-------|
| Oracle                         | -0.429 | -0.264 | -0.043 | 0.269 |
| FA-Gaussian                    | -0.232 | -0.082 | 0.114  | 0.385 |
| EM-Gaussian                    | -0.237 | -0.086 | 0.112  | 0.385 |
| NL-FA                          | -0.276 | -0.127 | 0.073  | 0.353 |
| BLR                            | -0.178 | -0.056 | 0.125  | 0.390 |
| Ridge (complete case)          | -0.163 | -0.047 | 0.129  | 0.391 |
| Lifted cavity (ours)           | -0.324 | -0.163 | 0.052  | 0.353 |
| Lifted cavity, $K = 2$         | -0.320 | -0.159 | 0.056  | 0.357 |
| Lifted sites, no cavity        | -0.275 | -0.123 | 0.078  | 0.359 |
| Scalar sites ( $K = 0$ )       | -0.097 | 0.034  | 0.205  | 0.443 |
| Transformer                    | -0.295 | -0.133 | 0.081  | 0.379 |
| Transformer, all masks         | -0.297 | -0.134 | 0.080  | 0.367 |
| Residual MLP                   | -0.287 | -0.131 | 0.074  | 0.360 |
| Scalar-cavity network          | 0.047  | 0.161  | 0.309  | 0.516 |
| Lifted cavity transformer (v3) | -0.329 | -0.167 | 0.047  | 0.346 |

Table 18: Air Quality CO: mean test NLL for every number of simulated missing sensors (23 later weeks).

| Method                                                   | $k=0$  | $k=1$  | $k=2$  | $k=3$ |
|----------------------------------------------------------|--------|--------|--------|-------|
| EM-Gaussian                                              | 0.074  | 0.063  | 0.085  | 0.146 |
| FA-Gaussian                                              | 0.052  | 0.086  | 0.116  | 0.154 |
| NL-FA                                                    | 0.144  | 0.120  | 0.091  | 0.124 |
| Bayesian linear regression                               | 1.037  | 0.433  | 0.216  | 0.202 |
| Ridge (complete case)                                    | 0.824  | 0.337  | 0.208  | 0.210 |
| GP, complete case <sup>†</sup>                           | 3.364  | 1.056  | 0.503  | 0.146 |
| $k$ -nearest neighbours <sup>†</sup>                     | 0.571  | 0.391  | 0.353  | 0.274 |
| Lifted cavity, zero-shot                                 | 0.044  | 0.064  | 0.129  | 0.160 |
| Transformer, zero-shot                                   | -0.017 | -0.006 | 0.081  | 0.154 |
| <b>Lifted cavity, fine-tuned (ours)</b>                  | -0.040 | -0.031 | 0.008  | 0.077 |
| no synthetic pretraining                                 | -0.001 | -0.014 | -0.006 | 0.077 |
| no cavity input                                          | -0.086 | -0.065 | -0.028 | 0.038 |
| scalar sites ( $K = 0$ )                                 | -0.026 | -0.021 | -0.003 | 0.075 |
| Transformer, fine-tuned                                  | -0.163 | -0.149 | -0.052 | 0.036 |
| Residual MLP, fine-tuned                                 | 0.025  | 0.061  | 0.110  | 0.160 |
| <i>Lifted cavity transformer, fine-tuned<sup>‡</sup></i> | -0.099 | -0.054 | -0.027 | 0.044 |

Table 19: Air Quality NO<sub>2</sub>: mean test NLL for every number of simulated missing sensors (23 later weeks).

| Method                                                   | $k=0$  | $k=1$  | $k=2$  | $k=3$  |
|----------------------------------------------------------|--------|--------|--------|--------|
| EM-Gaussian                                              | -0.237 | -0.174 | -0.175 | -0.168 |
| FA-Gaussian                                              | -0.196 | -0.173 | -0.155 | -0.162 |
| NL-FA                                                    | -0.090 | -0.102 | -0.142 | -0.169 |
| Bayesian linear regression                               | -0.055 | -0.214 | -0.178 | -0.160 |
| Ridge (complete case)                                    | -0.200 | -0.230 | -0.176 | -0.145 |
| GP, complete case <sup>†</sup>                           | 25.217 | 0.696  | -0.198 | -0.258 |
| $k$ -nearest neighbours <sup>†</sup>                     | -0.010 | -0.020 | -0.063 | -0.128 |
| Lifted cavity, zero-shot                                 | -0.350 | -0.295 | -0.233 | -0.176 |
| Transformer, zero-shot                                   | -0.360 | -0.310 | -0.231 | -0.136 |
| <b>Lifted cavity, fine-tuned (ours)</b>                  | -0.412 | -0.366 | -0.337 | -0.273 |
| no synthetic pretraining                                 | -0.419 | -0.365 | -0.316 | -0.248 |
| no cavity input                                          | -0.423 | -0.375 | -0.332 | -0.275 |
| scalar sites ( $K = 0$ )                                 | -0.359 | -0.337 | -0.302 | -0.231 |
| Transformer, fine-tuned                                  | -0.485 | -0.429 | -0.381 | -0.294 |
| Residual MLP, fine-tuned                                 | -0.246 | -0.198 | -0.147 | -0.106 |
| <i>Lifted cavity transformer, fine-tuned<sup>‡</sup></i> | -0.404 | -0.396 | -0.359 | -0.292 |

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